

Diffusion and Flow-Based Models for Controllable Image Generation

Abstract

In the rapidly advancing realm of visual generation, denoising diffusion probabilistic models and flow matching frameworks have fundamentally revolutionized the landscape of artificial intelligence. By establishing a mathematically rigorous bridge between non-equilibrium thermodynamics, continuous normalizing flows, and optimal transport, these models have marked a significant shift in generative capabilities, achieving unprecedented fidelity in text-guided image synthesis. However, relying solely on text for conditioning these models does not fully cater to the varied and complex requirements of real-world applications. Text prompts are often semantically ambiguous, making them inadequate for specifying precise spatial layouts, preserving the structural integrity of source images, or maintaining the strict identity of individual subjects across varying contexts. Acknowledging this critical shortfall, an extensive body of literature has emerged focusing on controllable generation with text-to-image (T2I) diffusion and flow-based models.

This survey provides an exhaustive, long-form academic review of both the theoretical foundations and the practical advancements in controllable visual generation. The analysis begins by unifying diffusion models and flow matching under the framework of probability flow ordinary differential equations (ODEs), exploring the mechanistic transition from curved, stochastic sampling trajectories to straight, deterministic rectified flows. The report subsequently categorizes the expansive landscape of controllable generation into four coherent method families: training-free guidance and attention manipulation, spatial and structural conditioning adapters, inversion-based semantic image editing, and subject-driven personalization. Through this comprehensive taxonomy, the report examines how techniques such as classifier-free guidance, optimal control, and zero-initialized neural adapters negotiate the critical trade-off between prompt adherence and visual fidelity. Furthermore, the survey identifies prevailing computational bottlenecks, evaluates novel metrics for generative consistency, and discusses the future trajectory of the field toward unified, vision-centric multimodal generative frameworks.

Introduction

Deep learning-based generative models have evolved at an extraordinary pace over the past decade, driven by the escalating demand for high-dimensional data synthesis in applications ranging from digital art creation to physical sciences. Early generative paradigms, notably Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), established the foundational principles for efficient image generation [1, 2]. GANs enabled real-time, high-fidelity synthesis through adversarial training but historically suffered from severe training instability and mode collapse. Conversely, VAEs provided stable optimization and explicit latent representations but often yielded blurry outputs due to the limitations of their surrogate loss

functions [3].

The introduction of Denoising Diffusion Probabilistic Models (DDPMs) initiated a profound paradigm shift [4]. By formulating generation as a reverse Markovian process that systematically denoises data from a standard Gaussian distribution, diffusion models bypassed the adversarial training bottleneck, achieving an unprecedented balance between sample quality and distributional coverage [5, 6]. The subsequent development of Latent Diffusion Models (LDMs)—which perform the denoising process within a heavily compressed autoencoder latent space—drastically reduced computational complexity and democratized access to high-resolution T2I synthesis, culminating in widely adopted foundation models such as Stable Diffusion [7].

Despite these remarkable generative priors, contemporary T2I models remain fundamentally limited by the inherent ambiguity of natural language. As these models are increasingly deployed for structured, professional creation—encompassing architectural rendering, virtual try-on, cinematic video generation, and customized marketing—visual quality alone is insufficient [8, 9]. Users require granular control over the generative process. An unconditional or purely text-conditioned model frequently exhibits catastrophic neglect of specific prompt tokens, violates physical constraints, breaks compositional relationships, and fails to maintain identity persistence across different viewpoints and temporal frames [10, 11]. Consequently, the task of "controllable generation" has emerged as a primary focus of the generative AI community, aiming to steer diffusion priors using novel constraints such as depth maps, segmentation masks, bounding boxes, and reference images [12, 13].

Concurrently with the explosion of controllable generation techniques, the underlying mathematical framework of these models has undergone a critical evolution. While traditional diffusion models rely on reversing stochastic differential equations (SDEs), which results in highly curved sampling trajectories, recent breakthroughs in Flow Matching (FM) and Rectified Flow (RF) have reframed generative modeling as a deterministic optimal transport problem [14, 15, 16]. By straightening the probability paths from noise to data, flow-based models dramatically reduce the requisite number of function evaluations (NFEs), improve mathematical tractability, and enable near-exact latent inversion [17, 18, 19].

This survey presents an exhaustive and systematic analysis of the state-of-the-art in controllable image generation, focusing on both diffusion and flow-matching paradigms. The structure of this report is organized as follows: The Methods section first establishes the mathematical and theoretical foundations uniting diffusion and flow models. It then categorizes the literature into four distinct research directions: (1) training-free guidance and optimal control, (2) spatial, structural, and layout conditioning, (3) inversion and semantic image editing, and (4) subject-driven personalization and identity preservation. Following this, the Challenges and Prospects section critically evaluates current metrics, computational bottlenecks, safety implications, and the integration of biological and tabular modalities. The report concludes by synthesizing the trajectory of these technologies toward fully unified, vision-centric generative systems.

Methods

Theoretical Foundations: From Diffusion to Flow Matching

To systematically understand controllable generation, it is imperative to dissect the mathematical frameworks that govern these models. Both diffusion and flow-matching models operate on the principle of continuous-time density evolution, transforming a tractable base distribution into a complex target data distribution.

Diffusion Models and Score-Based SDEs

Diffusion models define a forward continuous-time process that systematically corrupts a data sample $\mathbf{x}_0 \sim p_{data}(\mathbf{x})$ into standard Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ over a time horizon $t \in [0, T]$. This corruption is modeled as a stochastic differential equation (SDE):

$$d\mathbf{x} = f(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

where $f(\mathbf{x}, t)$ is the linear drift coefficient, $g(t)$ is the diffusion coefficient representing the noise schedule, and $\mathbf{w}$ is standard Brownian motion [20]. Generating novel samples corresponds to simulating the reverse-time SDE, which necessitates knowledge of the marginal score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$. Through denoising score matching, a neural network $\mathbf{s}_{\theta}(\mathbf{x}, t)$ is trained to approximate this score [20, 21].

A fundamental property of this formulation is that for every forward SDE, there exists a corresponding deterministic probability flow ordinary differential equation (ODE) that induces the exact same marginal distributions $p_t(\mathbf{x})$ at all times [20, 22]. The probability flow ODE is given by:

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

This ODE formulation theoretically connects diffusion models to Continuous Normalizing Flows (CNFs), enabling deterministic generation and exact log-likelihood computation. However, because the forward SDE trajectories are intrinsically stochastic and highly curved, the resulting ODE vector field is non-linear and mathematically complex. Consequently, synthesizing high-fidelity images requires sophisticated numerical solvers and numerous discretization steps to prevent severe truncation errors [15, 23].

Flow Matching and Rectified Flow

Flow matching has emerged as a generalized, simulation-free paradigm that trains CNFs

without the need to simulate forward SDEs or explicitly evaluate intractable marginal vector fields [14, 24]. Instead of learning to reverse a specific noise schedule, flow matching directly regresses a time-dependent vector field $v_\theta(\mathbf{x}, t)$ to match a target velocity field $u_t(\mathbf{x})$ that transports a base noise distribution p_0 to the data distribution p_1 [25].

Because the marginal vector field $u_t(\mathbf{x})$ is generally intractable for arbitrary data distributions, the Conditional Flow Matching (CFM) objective relies on constructing per-sample conditional probability paths $p_t(\mathbf{x}|\mathbf{x}_1)$ and their corresponding conditional vector fields $u_t(\mathbf{x}|\mathbf{x}_1)$. The CFM loss is elegantly defined as:

$$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t \sim U(0,1), \mathbf{x}_1 \sim p_1, \mathbf{x} \sim p_t(\mathbf{x}|\mathbf{x}_1)} \|v_\theta(\mathbf{x}, t) - u_t(\mathbf{x}|\mathbf{x}_1)\|^2$$

A profound specialization of this framework is Rectified Flow (or Optimal Transport CFM), which constructs the conditional paths as straight lines connecting the noise $\mathbf{x}_0$ and the data $\mathbf{x}_1$ [15, 26, 27]. The linear interpolant is defined as $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$. Under this formulation, the constant conditional velocity is simply $u_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathbf{x}_1 - \mathbf{x}_0$. By minimizing the mean squared error between $v_\theta(\mathbf{x}_t, t)$ and $(\mathbf{x}_1 - \mathbf{x}_0)$, the neural network learns an empirical barycentric projection of the optimal transport map [15, 28]. Rectified flows generate nearly straight ODE trajectories, drastically reducing discretization errors. This permits high-fidelity image generation with extremely few integration steps [29, 30]. Iterative "reflow" procedures—retraining the model on its own generated noise-data pairs—progressively straighten the learned ODE paths, establishing the mathematical foundation for state-of-the-art architectures like Stable Diffusion 3 and FLUX [31, 32].

Architectural Shift: Multi-Modal Diffusion Transformers (MM-DiT)

The theoretical shift from diffusion to flow matching has been accompanied by a structural shift from convolutional U-Nets to Diffusion Transformers (DiTs) [33, 34]. DiTs process flattened image patches as sequences, leveraging the scalability and global receptive field of transformer blocks. A critical innovation within DiTs is Adaptive Layer Normalization (AdaLN),

which injects the continuous timestep t and conditioning signals directly into the scale and shift parameters of the normalization layers [33]. This allows the network to dynamically and globally modulate feature representations across different integration steps. Recent investigations into DiT frequency domains, such as Frequency-Aware Flow Matching (FreqFlow), reveal that flow matching inherently generates low-frequency global structures in early stages and refines high-frequency textures later [35, 36]. Exploiting this frequency bias is central to advanced controllable generation.

Feature	Diffusion Models (Score-Based)	Flow Matching / Rectified Flow
Generative Trajectory	Curved, derived from reversing stochastic noise.	Straight (in RF), defined by linear interpolation.
Training Objective	Denoising score matching (predicting noise/score).	Vector field regression (predicting velocity).
Sampling Efficiency	Low; requires many steps to mitigate truncation error.	High; straight paths allow few-step Euler solvers.
Inversion Quality	Poor; exact ODE inversion accumulates massive error.	Excellent; straight ODEs allow near-perfect bidirectional flow.

Training-Free Guidance and Optimal Control

Controlling pre-trained generative models without altering their underlying weights is a highly desirable paradigm, as it preserves the vast visual priors acquired during large-scale pre-training while minimizing computational overhead.

Classifier and Classifier-Free Guidance

Explicit conditioning originally manifested as Classifier Guidance, where an auxiliary,

noise-conditioned classifier $p_\phi(y|\mathbf{x}_t)$ perturbed the score estimate toward a target class y [37]. The modified score was defined as $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) + \omega \nabla_{\mathbf{x}} \log p_\phi(y|\mathbf{x}_t)$, with ω serving as the guidance scale. However, this required training classifiers for every conceivable condition and frequently induced adversarial artifacts [37].

Classifier-Free Guidance (CFG) resolved this dependency by jointly training a single diffusion model on both conditioned and unconditioned distributions. During inference, the velocity or score is extrapolated away from the unconditional prediction:

$$\tilde{v}_\theta(\mathbf{x}_t, c, t) = v_\theta(\mathbf{x}_t, \emptyset, t) + \omega (v_\theta(\mathbf{x}_t, c, t) - v_\theta(\mathbf{x}_t, \emptyset, t))$$

While CFG vastly improves prompt adherence, it introduces a significant trade-off: high guidance scales push the latent samples off the natural data manifold, causing over-saturation, color shifting, and structural collapse [38, 39]. In the context of flow matching, standard CFG is

particularly problematic in the early stages of generation (near $t = 0$), where velocity

estimates are highly uncertain, misdirecting the ODE trajectory entirely [40]. To address this, CFG-Zero* optimizes a dynamic scaling parameter and enforces a zero-initialization scheme for the ODE solver, significantly mitigating trajectory inaccuracies when the model is under-fitted [40]. Furthermore, Manifold Projection techniques interpret CFG through the lens of homotopy optimization. By reformulating CFG sampling with an explicit manifold constraint, and utilizing incremental gradient descent with Anderson Acceleration, Manifold Projection ensures the extrapolated vector remains tangent to the smoothed data manifold, preserving extreme fidelity even at massive guidance scales [41, 42].

Energy-Based and Source-Guided Flow

In flow matching, classical guidance has been generalized using Optimal Control theory. Frameworks such as OC-Flow and Source-Guided Flow Matching (SGFM) reframe controllable

generation as minimizing a target energy function $R(\mathbf{x}_1)$ while adhering closely to the base velocity field [43, 44]. Rather than altering the pre-trained vector field directly—which risks

moving off-manifold—SGFM modifies the source noise distribution p_0 itself. This reduces the guidance problem to sampling from a biased source distribution, theoretically guaranteeing that the exact target distribution is recovered without retraining [44]. Similarly, Variational Flow Matching (VFM) casts controlled generation as a Bayesian inference problem, utilizing Monte Carlo sampling to estimate asymptotically exact guidance fields for unconditional models [45].

Attention Manipulation and Bounded Control

The cross-attention layers of diffusion models serve as the primary interface between text embeddings and spatial image features. Training-free layout control relies heavily on manipulating these matrices. Techniques like Attend-and-Excite and BoxDiff manipulate cross-attention maps during the reverse diffusion process, computing loss functions that force specific text tokens to maximally activate within user-defined bounding boxes [46, 47, 48]. However, multi-subject generation often suffers from "semantic leakage," where visual features of one subject bleed into another due to the global nature of attention mechanisms [49]. Bounded Attention mitigates this by restricting information flow within both self-attention and cross-attention matrices, explicitly masking out interactions between tokens that belong to distinct spatial regions to ensure each subject "remains itself" [49]. In MM-DiT architectures, Dual-Channel Attention Guidance (DCAG) separates the manipulation of the Key channel (which non-linearly determines *where* to attend via softmax) and the Value channel (which linearly aggregates *what* features to pull). Theoretical analysis reveals that controlling the Key channel provides coarse, dominant layout routing, while the Value channel facilitates fine-grained semantic injection [50].

Spatial, Structural, and Layout Conditioning

When training-free attention manipulation proves insufficient for dense structural control (e.g., depth maps, edge detection, human pose skeletons), explicit architectural adapters must be introduced to inject spatial priors into frozen backbone models.

Auxiliary Adapter Networks

ControlNet represents a watershed development in spatial conditioning [51]. It operates by cloning the encoding layers of a pre-trained U-Net or DiT and passing the cloned features through "zero-convolutions"—convolutional layers initialized with zero weights. This initialization guarantees that, at the commencement of training, the ControlNet branch outputs strictly zeros, leaving the robust, pre-trained generative prior entirely unaffected. As training progresses, the zero-convolutions gradually learn to integrate structural maps (e.g., Canny edges, depth maps) into the generative process without inducing catastrophic forgetting of the base model's capabilities [51, 52].

Similarly, the T2I-Adapter utilizes a lightweight, decoupled network to extract features from condition images, injecting them directly into the intermediate layers of the diffusion model [53]. Because adapters operate additively, multiple T2I-Adapters can be composed to simultaneously condition the generation on edges, color palettes, and depth. To optimize the considerable computational overhead of running parallel network branches, architectures like FlexControl introduce differentiable routers. These routers dynamically select which transformer blocks to activate based on the timestep and specific task, drastically accelerating inference without degrading structural alignment [54].

Region-Based and Layout-to-Image (L2I) Generation

While ControlNet excels at dense, pixel-aligned control, layout-to-image (L2I) generation requires the model to interpret sparse bounding boxes and category labels. GLIGEN addresses this by inserting new, trainable gated self-attention layers into the frozen diffusion backbone, directly grounding visual features to corresponding spatial coordinates provided by the layout [55]. Alternatively, LayoutDiffusion fuses multi-resolution image patches with layout coordinates into a unified structural representation, preventing the spatial ambiguity that arises from treating bounding boxes as mere text tokens [56].

To manage highly complex scenes with numerous overlapping entities, divide-and-conquer strategies such as DivCon decouple L2I generation into sequential sub-tasks. The model first reasons over numerical and spatial logic using a Large Language Model (LLM) to generate an optimal layout, and subsequently employs an iterative diffusion process to synthesize objects progressively from easiest to most complex [57]. Hierarchical Controllable (HiCo) models further disentangle spatial layouts through multiple weight-sharing branch networks, aggregating foreground and background features independently to prevent the common failure mode of object omission in dense semantic scenes [58].

Inversion and Semantic Image Editing

Text-guided image editing requires modifying specific attributes of a source image while strictly preserving identity-irrelevant background and structural information. The foundational step for real-image editing is inversion: mathematically mapping the image back into the model's latent noise space.

Diffusion vs. Flow-Based Inversion

In traditional DDPMs, DDIM inversion is a deterministic process that integrates the probability flow ODE backward in time. However, due to the severe non-linear curvature of the diffusion ODE, truncation errors accumulate rapidly. Consequently, forward generation from the

inverted noise vector often fails to perfectly reconstruct the source image [59, 60]. Null-text inversion mitigates this by optimizing the unconditional text embedding $\emptyset$ at each timestep during reconstruction, ensuring that the CFG trajectory perfectly matches the inversion trajectory without altering the base model weights [61]. Direct Inversion and Consistent-Inversion bypass this heavy computational cost by checking whether an intermediate target trajectory can be reversed toward the source trajectory, utilizing the discrepancy to correct the generative path on the fly [62, 63].

With the advent of flow matching, inversion has fundamentally shifted. Because rectified flows are explicitly trained to follow straight paths, their ODE inversion is intrinsically more accurate, requiring no test-time optimization [17, 64]. Piecewise Rectified Flow Inversion (PerRFI) and Bidirectional Flow Matching (BiFM) jointly learn generation and inversion within a single unified velocity field. BiFM predicts bidirectional average velocities, constrained by MeanFlow identities, allowing for near-perfect image reconstruction and semantic editing in as few as 4 to 8 steps [65, 66].

Semantic Editing Frameworks

Once a robust inversion is achieved, the editing phase manipulates the latent representations to reflect the target prompt.

- Attention-Based Editing:** Prompt-to-Prompt replaces the cross-attention maps of the target prompt with those generated by the source prompt. This forces the structural layout of the generated image to mirror the original, while allowing the target text to alter semantic textures [67]. MasaCtrl extends this concept to real images by converting self-attention into mutual self-attention, querying structural features directly from the source image during the synthesis of the edited image to maintain rigid spatial consistency [68].
- Flow-Based Velocity Steering:** FlowChef and FlowBypass steer the generative process entirely without costly backpropagation [69, 70]. FlowBypass analytically constructs a bypass trajectory directly connecting the inversion and reconstruction paths, avoiding the error accumulation typical of full inversions [70]. Kalman-Edit formulates the rectified flow editing process as a Linear Quadratic Gaussian (LQG) optimal control problem, utilizing a Kalman filter to dynamically estimate and suppress irrelevant semantic drift in background regions [71]. FluxSpace identifies a linearly interpretable semantic representation space within MM-DiT blocks, enabling disentangled, domain-agnostic edits (e.g., changing the season of a landscape or the species of an animal) simply by shifting vectors in the representation space, entirely circumventing the need for manual masking [72].

Methodology Family	Representative Models	Core Mechanism	Primary Application
Attention Manipulation	Prompt-to-Prompt, MasaCtrl, BoxDiff	Direct overriding of cross/self-attention	Localized texture changes, strict

		maps during reverse synthesis.	layout preservation.
Latent Optimization	Null-text Inversion, DragDiffusion	Test-time gradient descent on unconditional embeddings or latent vectors.	High-fidelity real-image inversion and point-based manipulation.
Velocity/ODE Steering	FlowChef, FlowBypass, KalmanEdit	Deterministic modification of the flow matching vector field / bypass trajectories.	Fast, inversion-free semantic transformation in flow models.
Structural Adapters	ControlNet, T2I-Adapter, GLIGEN	Injection of dense/sparse conditions via zero-initialized auxiliary networks.	Edge-guided, depth-guided, and bounding-box driven synthesis.

Subject-Driven Personalization and Identity Preservation

Subject-driven text-to-image personalization aims to capture the specific, unique identity of a user-provided concept (e.g., a specific person's face, a unique commercial product) from a few reference images and synthesize it seamlessly in novel contexts [73, 74].

Optimization-Based Personalization

Early approaches required computationally intensive per-subject fine-tuning. Textual Inversion

optimizes a novel pseudo-word embedding (e.g., S^*) in the text encoder's latent space to represent the subject, keeping the diffusion model weights entirely frozen [75]. Conversely, DreamBooth fine-tunes the entire U-Net alongside a class-specific prior preservation loss. This loss prevents the model from experiencing catastrophic forgetting regarding the general semantic class of the subject (e.g., forgetting what a generic "dog" looks like when fine-tuning on a specific pet) [76, 77]. While capable of achieving exceptionally high fidelity, these methods are prone to overfitting, severely hindering real-time deployment and rapid iteration [74].

Tuning-Free Adapters and Embeddings

To bypass test-time optimization, modern frameworks utilize pretrained encoders to inject identity features dynamically. The IP-Adapter elegantly decouples the cross-attention mechanism into a text-conditional branch and an image-conditional branch, allowing it to

ingest CLIP image embeddings directly as visual prompts [78].

For highly sensitive domains such as human face personalization, PhotoMaker and InstantID extract detailed facial representations (including facial landmarks and robust ID embeddings) and project them into the generative stream. InstantID integrates an IP-Adapter with a specialized ControlNet conditioned purely on facial landmarks to enforce strict structural identity without overriding style [79, 80]. UniID represents a synergistic evolution, combining both text embedding space injection and cross-attention injection. During inference, UniID employs a normalized rescaling strategy that allows the identity signals from both branches to reinforce each other without overwhelming the text-control capabilities of the base diffusion model, achieving superior prompt adherence [81].

Identity Preservation in Video (IPT2V)

Extending personalization to Identity-Preserving Text-to-Video (IPT2V) generation introduces severe temporal constraints. Models must preserve high-frequency intrinsic identity markers (e.g., specific facial features) while accommodating low-frequency global changes such as head rotation, lighting variations, and dynamic expressions [82, 83]. ConsisID addresses this by leveraging a frequency-aware heuristic control scheme within a DiT backbone, explicitly decomposing spatial features into distinct frequency bands to maintain identity persistence over time [82].

MoCA (Mixture of Cross-Attention) integrates Hierarchical Temporal Pooling and Temporal-Aware Cross-Attention experts directly into the DiT blocks. By capturing facial cues across varying temporal timescales, MoCA effectively mitigates identity drift during large motion variations and scene transitions [84]. Furthermore, spatial-temporal decoupled frameworks resolve the inherent conflict between spatial layout fidelity and motion smoothness by optimizing semantic structures and motion dynamics in isolated, sequential stages [85].

Challenges and Prospects

While the transition from stochastic diffusion to deterministic flow matching has fundamentally enhanced controllable image generation, several critical challenges remain unresolved, dictating the trajectory of future research.

Evaluation Metrics and Generative Consistency

Evaluating controllable generation is inherently difficult due to the subjective nature of aesthetic quality and prompt alignment. Standard metrics like the Fréchet Inception Distance (FID) and CLIP scores frequently fail to capture localized structural adherence or subtle identity degradation [86, 87]. To address this, Flow-based Likelihood Distance (FLD) leverages the exact density estimation properties of normalizing flows to compute how closely generated images align with the true distribution of a specific domain, providing a more rigorous mathematical assessment of realism than feature-space approximations [87]. Furthermore, generative consistency is increasingly recognized as a multi-dimensional target—encompassing prompt faithfulness, identity persistence, multi-view coherence, and physical plausibility—necessitating unified, overlap-aware benchmark suites rather than isolated task metrics [8, 88].

Computational Bottlenecks and Edge Deployment

Despite the sampling acceleration offered by rectified flows, massive multi-modal transformers containing billions of parameters still pose a formidable barrier for deployment on resource-constrained edge devices. Strategies such as progressive step distillation and ultra-low-bit architectural pruning (e.g., W8A8 quantization) are actively researched to compress DiT architectures [89, 90].

A novel approach to this bottleneck is Warm-Start Flow Matching (WS-FM). WS-FM proposes a paradigm where computationally lightweight generative models produce low-quality draft samples, which are subsequently utilized as the initial distribution for the FM algorithm closer to

the terminal time $t = 1$. Unlike traditional generation that starts from pure noise at $t = 0$, this acts as a learning-to-refine process, providing a mathematically guaranteed speed-up factor without degrading final sample quality [91].

Safety, Decontextualization, and Defense Mechanisms

As identity preservation and deepfake methods become virtually indistinguishable from reality, unauthorized mimicry poses severe ethical and legal risks. Frameworks like DeContext operate as an in-context defense mechanism, identifying the multimodal attention pathways within DiTs that propagate recognizable identity information and proactively disrupting them. By regularizing cross-attention during generation, DeContext prevents unauthorized identity impersonation while maintaining the overall aesthetic quality and instruction-following capability of the edited image [92].

Conversely, tracing unauthorized training data within personalized models is achieved through techniques like SIREN. SIREN embeds proactive, imperceptible coatings into source images prior to online release. If a personalized model is fine-tuned on this coated data, the latent feature space alignment forces the model to exhibit a highly traceable watermark upon generation, ensuring copyright compliance and data provenance [93].

Toward Vision-Centric Multimodal Flows

Current controllable generation pipelines remain highly fragmented, requiring separate, manually engineered adapters for depth, layout, and text. The next frontier involves unifying these disparate modalities into purely visual flows. Architectures such as FlowInOne render all textual instructions, layout bounding boxes, and structural constraints directly as visual prompts onto a single input canvas. The generative model then learns a continuous visual evolution from the input canvas to the target output using a single, unified flow matching objective. This elegant formulation eliminates the bottleneck of cross-modal alignment and task-specific architectural branches, signaling a definitive transition toward fully vision-centric foundation models where perception, reasoning, and creation are seamlessly integrated within a single deterministic flow [94].

Conclusion

The landscape of controllable image generation has undergone a profound and rapid evolution, transitioning from the stochastic approximation of diffusion models to the deterministic,

geometrically optimized trajectories of flow matching and rectified flows. This mathematical maturation has been strictly paralleled by architectural shifts toward Diffusion Transformers, unlocking unprecedented capabilities in model scaling and dynamic feature modulation. The development of training-free guidance mechanisms, zero-initialized spatial adapters, and bidirectional ODE inversion techniques has provided practitioners with granular control over spatial layouts, semantic edits, and subject identity without corrupting pre-trained visual priors. As the field looks forward, addressing the inherent trade-offs between strict structural adherence and generative diversity remains paramount. The ongoing integration of optimal control theory, unified multimodal flow formulations, and rigorous safety alignment frameworks promises to resolve current computational bottlenecks and ethical concerns. Ultimately, the convergence of flow-based exact density estimation and sophisticated attention control mechanisms is set to yield generative systems that are not merely capable of synthesizing high-fidelity pixels, but inherently capable of deep, physically and semantically consistent visual reasoning.

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